

```
/*  
 * C Program to Perform Binary Search using Recursion  
 */
```

```
#include <stdio.h>
```

```
void binary_search(int [], int, int, int);
```

```
void bubble_sort(int [], int);
```

```
int main()
```

```
{
```

```
    int key, size, i;
```

```
    int list[25];
```

```
    printf("Enter size of a list: ");
```

```
    scanf("%d", &size);
```

```
    printf("Enter elements\n");
```

```
    for(i = 0; i < size; i++)
```

```
    {
```

```
        scanf("%d",&list[i]);
```

```
    }
```

```
    bubble_sort(list, size);
```

```
    printf("\n");
```

```
    printf("Enter key to search\n");
```

```
    scanf("%d", &key);
```

```
binary_search(list, 0, size, key);
```

```
}
```

```
void bubble_sort(int list[], int size)
```

```
{
```

```
    int temp, i, j;
```

```
    for (i = 0; i < size; i++)
```

```
    {
```

```
        for (j = i; j < size; j++)
```

```
        {
```

```
            if (list[i] > list[j])
```

```
            {
```

```
                temp = list[i];
```

```
                list[i] = list[j];
```

```
                list[j] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
void binary_search(int list[], int lo, int hi, int key)
```

```
{
```

```
    int mid;
```

```
if (lo > hi)
{
    printf("Key not found\n");
    return;
}
mid = (lo + hi) / 2;
if (list[mid] == key)
{
    printf("Key found\n");
}
else if (list[mid] > key)
{
    binary_search(list, lo, mid - 1, key);
}
else if (list[mid] < key)
{
    binary_search(list, mid + 1, hi, key);
}
}
```